

NUCLEUS

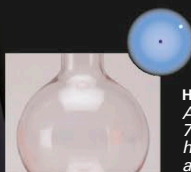
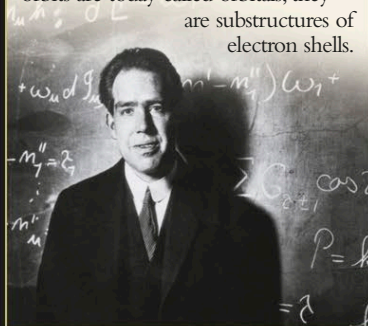
A tightly bound ball of six protons (purple) and six neutrons (gold)

CHEMICAL ELEMENTS

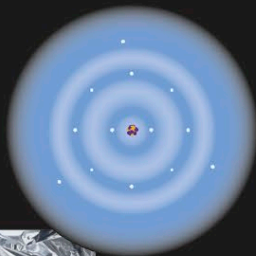
Atoms are not all the same—they can hold different numbers of protons, neutrons, and electrons. A substance made of atoms of just one type is called a chemical element and is given an atomic number equal to the number of protons, and thus electrons, in its atoms. Examples are hydrogen, with an atomic number of 1 (all hydrogen atoms contain one proton and one electron), helium (atomic number 2), and carbon (number 6). Altogether, there are 94 naturally occurring elements. The atoms of any element are all the same size and, crucially, contain the same configuration of electrons, which is unique to that element and gives it specific chemical properties. The universe once consisted almost entirely of the lightest elements, hydrogen and helium. Most of the others, including such common ones as oxygen, carbon, and iron, have largely been created in stars and star explosions.

NIELS BOHR

Danish physicist Niels Bohr (1885–1962) was the first to propose that electrons in an atom move within discrete “orbits.” He suggested that these orbits have fixed energy levels and that atoms emit or absorb energy in fixed amounts (“quanta”) as electrons move between orbits. Bohr’s orbits are today called orbitals; they are substructures of electron shells.



HYDROGEN
A colorless gas at 70°F (21°C). Its atoms have just 1 proton and 1 electron in a single shell.



PROPERTIES OF ELEMENTS

Elements vary markedly in their properties, as shown by the four examples here. These properties are determined by the elements’ different atomic structures.



ALUMINUM
A solid metal at 70°F (21°C). Its atoms have 13 protons, 14 neutrons, and 13 electrons in 3 shells.



SULFUR
A yellow, brittle solid at 70°F (21°C). Its atoms have 16 protons, 16–18 neutrons, and 16 electrons in 3 shells.



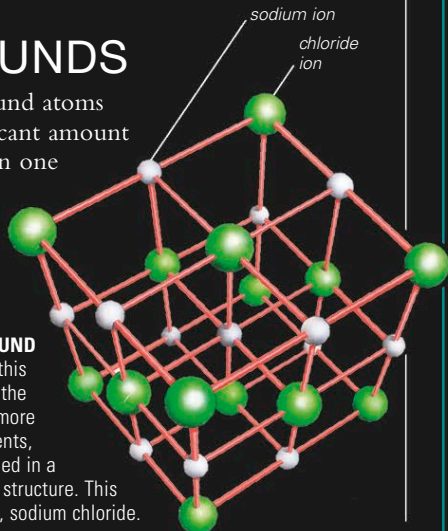
BROMINE
A fuming brown liquid at 70°F (21°C). Its atoms have 35 protons, 44 or 46 neutrons, and 35 electrons in 4 shells.

CHEMICAL COMPOUNDS

Most matter in the universe consists of unbound atoms or ions of a few chemical elements, but a significant amount exists as compounds, containing atoms of more than one element joined by chemical bonds. Compounds occur in objects such as planets and asteroids, in living organisms, and in the interstellar medium. In ionic compounds, such as salts, some atoms have gained and others have lost electrons. The resulting charged ions are typically (in salt crystals, for example) bonded by electrostatic forces and arranged in a rigid, repeating structure. In covalent compounds, such as water, the atoms are held in structures called molecules by the sharing of electrons between them.

IONIC COMPOUND

Compounds of this type consist of the ions of two or more chemical elements, typically arranged in a repeating solid structure. This example is salt, sodium chloride.



ELECTRON

Electrons have a negative charge and a mass more than a thousand times smaller than a proton or neutron

INSIDE A NEUTRON

Protons and neutrons are each made of three quarks, bound by gluons. The quarks flip between “red,” “green,” and “blue” forms, but there is always one of each color.